

FBISE Maths Class 9th Guess Paper 2nd Annual Exam 2026

- ▶ **Most Important Chapters : 4,6,8,9,11**
- ▶ **For Rest of the chapters, Do only Past Paper MCQs & Short Questions**
- ▶ **MCQs of Exercises & Past Papers are mandatory to cover**
- ▶ **Watch my Playlist of Mathematics to secure full marks as I have covered all chapters Past and Model Papers MCQ + Short + Long Questions in each Video**
- ▶ **Download all Material : Past & Model Papers along with Solved Papers PDF & Chapter wise Solved Past Paper Questions**
- ▶ **Do Following Most Important Questions to Grab Full Marks (75/75) in Math Paper**

Chapter 1

(i) $\frac{16^{\frac{1}{5}} \cdot 16^{\frac{1}{4}}}{16^{\frac{-3}{10}}}$ (ii) $7^{\frac{-1}{3}} (7^{\frac{5}{3}} - 7^{\frac{4}{3}})$ (iii) $\frac{2^{\frac{2}{3}} \cdot 2^{\frac{1}{7}}}{2^{\frac{1}{2}}}$

(iv) $\frac{3^{\frac{-1}{2}} \cdot 3^{\frac{1}{2}}}{3^{\frac{1}{2}}}$ (v) $\left(\frac{36^{\frac{1}{2}} \cdot 6^{\frac{1}{2}}}{8^{\frac{1}{2}} \cdot 27^{\frac{1}{2}}} \right)^3$ (vi) $\left(\frac{2187 a^5 b^{17}}{a^{12} b^3} \right)^{\frac{1}{7}}$

(vii) $\sqrt[4]{\frac{a^3}{b^3}} \times \sqrt[4]{\frac{b^3}{c^3}} \times \sqrt[4]{\frac{c^3}{a^3}}$

5. Use suitable laws of exponents to show that

$$\left(\frac{x^p}{x^q} \right)^{p+q} \times \left(\frac{y^q}{y^r} \right)^{q+r} \times \left(\frac{z^r}{z^p} \right)^{r+p} \times x^{q^2} \times y^{r^2} \times z^{p^2} = x^{p^2} \times y^{q^2} \times z^{r^2}$$

4. Use the properties of exponents to simplify each of the following.

(i) $\frac{(-2)^3 \cdot (-2)^{-4} \cdot (-2)}{(-2)^{-3}}$ (ii) $\frac{2^{\frac{1}{2}} \cdot 2^{\frac{3}{4}} \cdot 3 \cdot 3^{\frac{3}{2}}}{2^{\frac{1}{2}} \cdot 3^{-\frac{1}{2}}}$

Chapter 2

Example 18: An earthquake that occurred in Pakistan in 2005, measured 7.6 on the Richter scale. In 1978, an earthquake in China measured 8.2 on the Richter scale. How many times, the China's earthquake was stronger than the Pakistan's earthquake.

3. The Kansu, China earthquake of 1920 was measured about 8.5 on Richter Scale and the Tokyo, Japan earthquake of 1923 was measured 7.8 on that scale how many times stronger was the 1920 earthquake than 1923 earthquake?

Chapter 3

Example 9:

If $U = \{1, 2, 3, \dots, 11\}$, $A = \{1, 2, 3, 5, 7, 9\}$, $B = \{1, 2, 3, 4, 5, 6\}$ and $C = \{5, 6, 7, 8\}$ then verify the associative property of intersection.

Example 5:

If $A = \{0, 1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$ and $C = \{6, 7, 8, 9\}$, find $A \cup (B \cap C)$ and $A \cap (B \cup C)$ through Venn diagram.

Example 6:

If $P = \{a, b, c, d, e\}$, $Q = \{f, g, h, i\}$, $R = \{d, e, f, g, j\}$
Find $(P \cup Q) \cap R$ and $(P \cap Q) \cup R$ through Venn diagram.

Example 7: If $P = \{3, 5, 7\}$, $Q = \{1, 3, 5, 7, 9\}$, $R = \{0, 2, 5, 8\}$, then find $P \cup (Q \cap R)$ using Venn diagram.

Example 8:

If $U = \{1, 2, 3, 4, 5, \dots, 15\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{1, 2, 3, 4, 5, 6, 7\}$ and $C = \{2, 3, 5, 7, 11\}$, then verify the associative property of union using Venn diagram.

Example 15:

Find the values of x and y when $(x - 3y, 5x + 1) = (4, 6)$

Example 16:

Exhibit $P \times Q$ and $Q \times P$ by arrow diagram when $P = \{a, b, c\}$ and $Q = \{d, e\}$.

Example 19:

Let $A = \{1, 2, 3\}$, $B = \{0, 1, 3\}$ and $R = \{(a, b) / a \in A, b \in B \text{ and } a > b\}$, then find R and show it by arrow diagram.

Example 20:

(a) If $A =$ Set of Natural numbers and $R = \{(x, y) | x \in A \wedge y \in A\}$ i.e. $R \subseteq A \times A$

Then find the domain and range of R .

Example 21:

Given that $A = \{0, 2, 3\}$, $B = \{0, 2, 4, 6, 9, 16\}$ and $R = \{(x, y) | x \in A \wedge y \in B \wedge x^2 = y\}$.

Verify: $\text{Dom } R^{-1} = \text{Range } R$ and $\text{Range } R^{-1} = \text{Dom } R$

Chapter 4

Example 3: Factorize $3x^2 + 22x - 16$

Example 1: Factorize the expression $8x^4 - 26x^2m^2 + 18m^4$.

Factorize: $(x^2 + 3x - 4)(x^2 + 3x + 5) + 8$

Factorize: $(x + 5)(x + 3)(x + 2)(x + 6) - 88$

Example 7: Factorize the following expressions.

(i) $27a^3 + 1 + 27a^2 + 9a$

(ii) $x^3 + \frac{3}{x} - \frac{1}{x^3} - 3x$

(iii) $4x^2(2x - 15) - 25(5 - 6x)$

Example 8: Factorize: $x^3 + 27y^3$

Example 9: Factorize $x^6 - y^6$.

Example 10: Factorize $8a^3 - 125b^3 - 2a + 5b$

Example 15: Find the HCF of $12 + 16a + 7a^2 + a^3$ and $13a + 5a^2 + 14 + a^3$.

Example 18: Find the LCM of $2x^2 - 12xy + 16y^2$, $x^2 - 6xy + 8y^2$ and $3x^2 - 12y^2$.

Example 19: Find the LCM of $P = 10x^4 + 3x^3 + 8$ and $Q = 8x^4 + 3x + 10$

$$\frac{1}{x(x-y)} + \frac{1}{y(x+y)}$$

Example 29: Find the square root of $9x^2 - 42xy + 49y^2$ by division method.

Example 30: Find the square root of $4\left(x^2 + \frac{1}{x^2}\right) + 12\left(x + \frac{1}{x}\right) + 17, x \neq 0$.

Example 31: To convert $x^4 + 4x^3 + 10x^2 + 10x + 5$ into a perfect square,
 i. What should be added to it?
 ii. What should be subtracted from it?
 iii. What should be the value of x ?

Example 32: For what values of m and n , $9x^4 - 24x^3 - 14x^2 + mx + n$ is a complete square or perfect square?

Chapter 5

Solve the following linear equations in one variable.

$$5. \quad \frac{x-2}{2} + \frac{x+10}{9} = 5$$

$$7. \quad \frac{x}{2} + \frac{x}{3} - \frac{x}{4} + \frac{x}{5} = 7\frac{5}{6}$$

Example 10: Solve the absolute value equation.

$$\frac{|10-x|}{5} = \frac{|2x-5|}{2}, \quad \text{where } x \in \mathbb{R}$$

Example 13: Solve the inequality.

$$3(-4 + 5y) \leq -8(1 - 2y) + 6, x \in \mathbb{R}$$

Example 14: Solve. $\frac{1}{2}x + 3 \geq \frac{1}{4}x + 2, x \in \mathbb{R}$

Chapter 6

Example 12:

Solve the following triangles. (Find unknown elements of the following triangles)

- i. $\triangle ABC, \angle C = 90^\circ, \angle A = 30^\circ, a = 6 \text{ cm}$
- ii. $\triangle ABC, \angle C = 90^\circ, c = 12 \text{ cm}, b = 6\sqrt{2} \text{ cm}$

Example 11:

Use basic trigonometric identities to prove the following relations.

- i. $\cos \alpha + \sin \alpha \tan \alpha = \sec \alpha$
- ii. $\frac{\tan \alpha}{\sec \alpha - 1} = \frac{\sec \alpha + 1}{\tan \alpha}$
- iii. $\frac{\sin^2 \alpha + 2 \cos \alpha - 1}{\sin^2 \alpha + 3 \cos \alpha - 3} = \frac{1}{1 - \sec \alpha}$

6.4.3 Pythagorean Identities

- i. $\sin^2 \theta + \cos^2 \theta = 1$
- ii. $\sec^2 \theta - \tan^2 \theta = 1$
- iii. $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

Proof:

1. Prove the following relations by using basic trigonometric identities.

ii. $\frac{1 - \sin \theta}{1 + \sin \theta} = (\sec \theta - \tan \theta)^2$

iii. $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{\sin \theta}{1 + \cos \theta}$

vi. $\frac{1}{\sec \theta - \tan \theta} = \sec \theta + \tan \theta$

7. Two men on opposite sides of a tree are in line with it. They observe that angles of elevation of top of tree are 30° and 60° . The height of tree is 15 m. Find distance between the men.
8. From the top of a tree, the angles of elevation and depression of the top and bottom of a building are 30° and 45° respectively. If height of tree is 12 m, find height of building and distance between tree and building.

Do All Bearing Questions from my Playlist Video on Past paper Questions

Chapter 7

Example 3:

Prove that A (3, 2), B (4, 4) and C (5, 6) are collinear points.

Example 4:

Prove that the points (3, 4), (3, 1) and (8, 4) are the vertices of a right triangle.

Example 6:

If (1, 4) is the midpoint of the line segment joining (6, b) and (a, 2). Find a and b.

Chapter 8

Example 1:

- (i) Find the gradient (slope) of line whose inclination is 45° .
- (ii) Find inclination of line whose gradient is 1.732.

Example 2:

Find gradients of lines joining: (i) A(1, 4), B(3, 6) (ii) C(-3, 5), D(0, 3)

Example 4:

Lines AB and CD with coordinates A(0, 3), B(2, 8), C(3, 10), D(8, k) are perpendicular. Find k.

Example 4:

Find the equation of straight line with the following information:

- (i) slope = 5, y-intercept = 3
- (ii) through (5, 0) with slope -3
- (iii) through (2, 4) and (-1, -4)
- (iv) x-intercept = 6, y-intercept = -4

Example 5:

Find the equation of straight line:

- (i) through (3, -2) and perpendicular to the line with slope 3.
- (ii) having y-intercept = -7 and parallel to the line with slope 4.
- (iii) through (-5, 1) and perpendicular to the line passing through (0, 9) and (-2, 6).
- (iv) if length of perpendicular from the origin is 6 units inclined at 60° .

Example 6:

Reduce the equation $6x - 5y + 15 = 0$ into:

- (i) slope-intercept form
- (ii) two-intercept form
- (iii) point-slope form
- (iv) two-point form
- (v) normal form
- (vi) symmetric form

Example 8:

Find the interior angles of triangle whose vertices are A(-2, 0), B(3, 0) and C(6, 5).

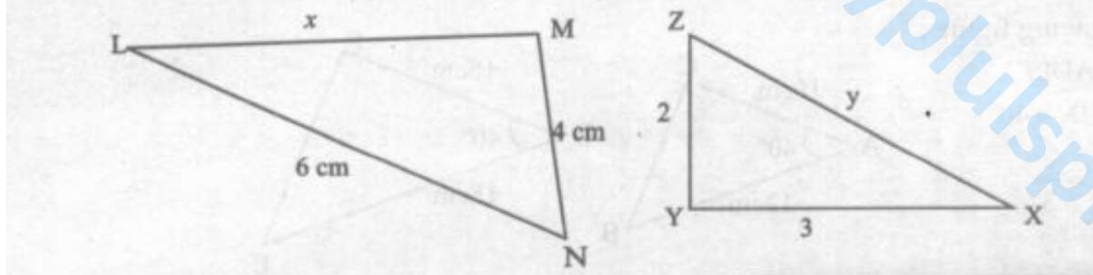
Example 10:

Find equation of line passing through point of intersection of lines $x + 2y = 0$, $x - y = 3$

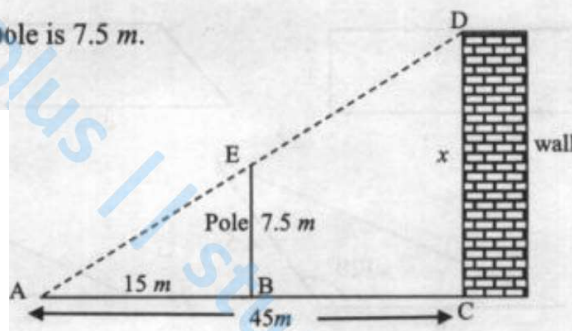
- (a) passing through point (1, 1)
- (b) parallel to $3x - 4y + 7 = 0$

Chapter 9

Example 5: In the figure below $\triangle LMN \sim \triangle XYZ$. Find values of x and y .



Example 6: In the figure height of pole is 7.5 m .
Find the height of wall.

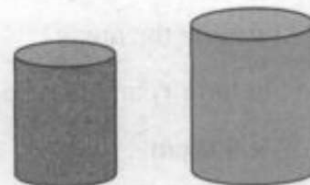


Example 9:

The ratio of the areas of the bases of two similar cylinders is $9 : 4$.

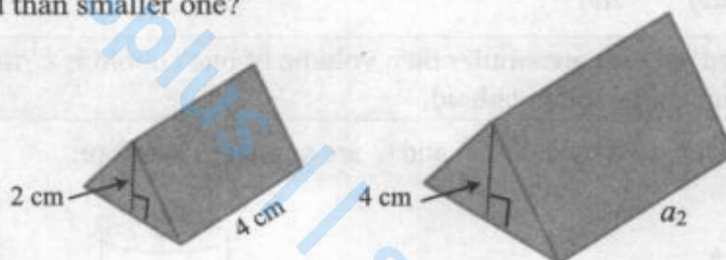
The area of the base of the smaller cylinder is 240 cm^2 .

- What is the area of base of larger cylinder?
- Write the ratio of heights of both cylinders.



Example 12:

Find a_2 if the following solids are similar. Also find the ratio of volumes of both. How many times is the volume of larger solid than smaller one?



Solution:

Example 13:

In the figure two geometrically similar cylinders are shown. Find:

- (i) the ratio of volume of smaller cylinder to larger cylinder.



- (ii) curved surface area of smaller cylinder if that of larger one is 250cm^2 .
 (iii) volume of larger cylinder if volume of smaller one is 162cm^3 .

Example 16:

Exterior angle of a regular polygon is 120° . Identify the name of polygon.

Example 17:

In a certain polygon, the sum of the measures of all the interior angles is equal to twice that of the exterior angles. What is the name of that polygon?

Example 18:

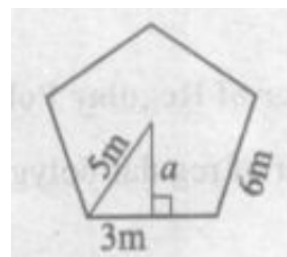
The measure of the exterior angles of a polygon are $(x + 4)^\circ$, $(3x - 4)^\circ$, $(7x - 3)^\circ$, $(2x + 3)^\circ$, $(8x - 1)^\circ$ and $(9x + 1)^\circ$.

- (i) Identify polygon. (ii) Find x . (iii) Find the measure of each angle.

Example 19:

Basement of a water tank is in the shape of regular pentagon having the side length 8 feet. Its apothem is 7 feet. Find the perimeter and area of basement of tank.

3. Base of a minaret of a Masjid is built in the shape of regular pentagon as shown in the figure. Find the perimeter and area of base of minaret.



For Locus Problems, watch my Chapter 9 Video

Chapter 10

Do all Exercise MCQs, Read End Chapter Key Points and Do Past Paper

MCQs

Draw angle bisectors, perpendicular bisectors, medians and altitudes of given triangle.

Chapter 11

Example 6: Find the arithmetic mean of the following data

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of students	2	5	7	3	8

Example 7:

The selling prices of the seven articles sold by a shopkeeper in rupees were 1075, 1050, 2050, 1030, 2045, 1090 and 1530. Find the median selling price of these articles.

Example 8:

Find the median of the following set of values.

40, 35, 45, 60, 50, 35, 40, 45, 35, 50

Example 9: Marks obtained by 100 students in an examination are given below,

Marks	10 – 19	20 – 29	30 – 39	40 – 49	50 – 59
No of students	5	25	40	20	10

Find the median marks.

Example 10: Find mode of following data sets.

(i) 30, 25, 40, 30, 45, 25, 30, 40, 30

(ii) 1, 2, 3, 2, 1, 3, 1, 2

(iii) 125, 130, 115, 125, 135, 120, 135, 130, 120

Example 11:

Fifty workers of a factory are distributed in 7 groups according to their wages in rupees per hour.

Wages in Rupees	25-29	30-34	35-39	40-44	45-49	50-54	55-59
No of workers	2	4	8	20	6	6	4

Find mode of the distribution.

Example 12:

Find median and mode for the following discrete frequency distribution.

x	1	2	3	4	5
f	3	4	5	2	2

Solution:

Example 14: The following are the monthly expenditure of a house according to their weights.

Items	Expenditure (Rs)	Weight relative importance
Food	8,000	20
House rent	5,000	10
Clothing	1,000	6
Utility bills	900	4
Education	600	2
Miscellanies	100	1

Find the weighted mean.

Example 20: A fair coin is tossed 150 times and head comes up 70 times. What is the relative frequency of getting head up.

Example 21: 5 students out of a section of 30 students are expected to be absent everyday in summer camp. What No. of students is expected to be absent every day from 12 such sections of same strength.

Example 18: There are 12 teachers in maths department among them 2 are higher secondary teachers, 4 are secondary teachers and 6 are senior elementary teachers. If good performance award is to be given to only one teacher then find probability

- That a secondary teacher will get the award
- That a secondary teacher will not get the award

Example 17: Ali picked a book at random from a library shelf which had 3 history books of grade 6, 4 geography books grade 7, 9 science books of grade 8, 3 science books of grade 7, 6 geography books of grade 8 and 5 science books of grade 6. Find the probability of getting:

- a science book
- a geography book
- a book of grade 7
- a book of grade 9
- a book of mathematics

Example 22: If Probability of a person to be Hafiz e Quran in a village is $\frac{3}{10}$. What do you expect the no. of Hafiz e Quran in 2500 such villages?

Do all Exercise MCQs, Past Paper MCQs & Past Papers Short & Long Questions- Best of Luck dear Students